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Thesis for Master of Public Policy and Social Change (MA)

Urban Climate Adaptation Governance in Nepal: Explaining the Persistence of Institutional Constraints in Local Governments

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Abstract

Urban local governments in Nepal hold formal constitutional authority over climate change adaptation yet remain institutionally constrained in exercising it. Despite two decades of policy learning, federal restructuring, and expanded formal mandates, municipalities in Kathmandu Valley continue to operate as implementers of centrally and donor-defined priorities rather than as autonomous drivers of adaptation governance. The thesis explains why these constraints persist by tracing the mechanisms through which barriers are reproduced across successive governance reforms, as a way to go beyond barrier identification. Through interpretive, single-case qualitative design, the study integrates three theoretical frameworks: historical institutionalism, multi-level governance, and the adaptive capacity wheel. Primary data from 18 semi-structured expert interviews and 6 qualitative questionnaire responses, conducted with municipal officials, national policymakers, international and non-governmental representatives, and researchers in Kathmandu Valley in 2025, are analyzed alongside national climate policy documents including the Local Governance Operation Act (2017), the National Climate Change Policy (2019), and the National Adaptation Plan (2021–2050).

The research makes three contributions. It extends historical institutionalism by showing that critical junctures in federal climate governance can be absorbed into path-dependent reproduction rather than generating transformation. It contributes to multi-level governance by showing that formal decentralization without fiscal, technical and administrative autonomy produces symbolic rather than genuine functional empowerment. Finally, it reconceptualizes adaptive capacity as a historically embedded governance arrangement that forms a mutually reinforcing governance trap which cannot be disrupted simply through capacity building, training, funding or policy layering.

Keywords: urban climate adaptation governance, institutional constraints, historical institutionalism, multi-level governance, adaptive capacity, federalism, Kathmandu, Nepal.

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Chapter 4 Methodology

The chapter outlines the research design of the study, organized according to the layers of *Saunders & Tosey (2012)* research onion (Figure 2): research philosophy, research approach, research strategy, data collection and analysis. The study employs a qualitative design centered on semi-structured expert interviews, supplemented by a qualitative questionnaire for stakeholders who could not be reached directly. The overall orientation is descriptive and exploratory, with an explicit explanatory aim: to understand why institutional constraints on urban climate adaptation in Nepal persist despite federalism, decentralization and expanded climate mandates. The empirical focus is urban climate adaptation governance in Kathmandu Valley, which serves as the primary case through which institutional dynamics are examined.

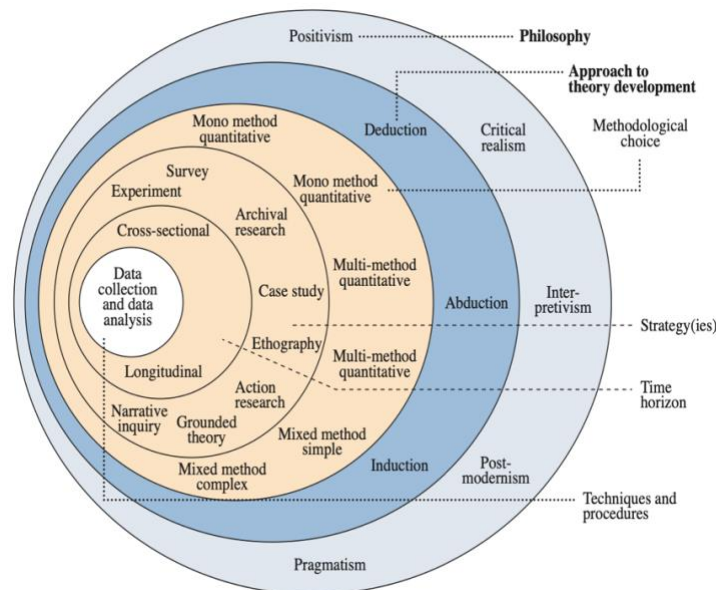


Figure 1 The research onion ¹ (Melnikovas, 2019)

¹ The research onion consists of six main layers: (Melnikovas, 2018)

1. *Research philosophy* – forms a basis of the research by delineation of *ontology* – nature of reality, *epistemology* – nature, sources of knowledge or facts and *axiology* – values, beliefs and ethics of the research.

2. *Approach to theory development* – can be implied by the research philosophy on previous level and usually include: *deduction* – the research starts with an existing theory, then rising a question or hypothesis and data collection in order to confirm or reject the hypothesis; *induction* – the research starts with observation and data collection, moving to description and analysis in order to form a theory; *abduction* – observation of an empirical phenomena is followed by the research which comes up with a best guess or conclusion based on available evidence. Deductive approach is applied for existing theory testing, while inductive approach is commonly used in developing a theory or in fields with little

4.1 Research Philosophy

4.1.1 Interpretivism

The study adopts an interpretivist research philosophy. Interpretivism as a philosophical paradigm assumes that social reality is not an objective, fixed structure waiting to be discovered, but is actively constructed by the actors who inhabit it (*Saunders & Tosey, 2012*)

“Interpretivism focuses on conducting research amongst people rather than upon objects, adopting an empathetic stance so as to understand their social world and the meaning they give to it from their point of view” (p.58)”.

Multiple, sometimes contradictory interpretations of the same phenomenon can coexist, because reality is shaped by actors’ subjective perceptions and social interactions (*Mummery & Mummery, 2019; Scauso, 2020*). For climate-change adaptation, understanding institutional constraints, power relations, and capacity gaps requires engaging with the meanings and lived experiences of those involved in governance, not just measuring variables from the outside. Ontologically, this approach treats institutions, authority relations and governance arrangements as socially produced and reproduced through practice. Epistemologically, it holds that valid knowledge about urban climate adaptation in Nepal is generated through close, contextual engagement with local actors rather than through hypothesis testing against a pre-defined model.

This stance is necessary because of the central research question of the study why institutional constraints persist despite formal reform which cannot be adequately answered through quantitative

research on the topic. Abductive approach usually starts with a surprising fact and is moving between induction and deduction in order to find the most likely explanation.

3. *Methodological choice* – determines the use of quantitative and qualitative methods or various mixtures of both.

4. *Strategy* – to collect and analyze data: experiment, survey, archival research, case study, ethnography, action research, grounded theory, narrative inquiry.

5. *Time horizons*. This layer defines the time frame for the research – *cross-sectional* or short-term study, involving collection of data at a specific point of time; *longitudinal* – collection of data repeatedly over a long period of time in order to compare data. 6. *Techniques and procedures* include data collection and analysis – the use of primary/secondary data, choosing sample groups, developing questionnaire content, preparing interviews, etc.

measurements alone. It requires understanding how actors experience, interpret and navigate those constraints within the governance system. Interpretivism also helps avoid the epistemic fallacy common in positivist vulnerability assessments, where social reality is reduced to what can be observed and statistically measured. Such approaches tend to focus on technical and economic interventions such as building higher bridges, flood barriers, drainage infrastructures while overlooking the deeper institutional, political and social conditions that determine a city's actual capacity to adapt to climate change (*Fletcher, 2024*). Within this paradigm, the robustness of the knowledge is therefore pursued through triangulation across multiple stakeholders such as municipal officials, planners, NGOs, and national-level actors and across multiple forms of qualitative data, rather than through methodological pluralism in a positivist sense.

4.2 Research approach

4.2.2. Deductive Approach

The study adopts a deductive research logic. The deductive element is grounded in the theoretical framework developed in Chapter 3, which draws on historical institutionalism, multi-level governance and the adaptive capacity wheel. These frameworks generate substantive expectations about how path-dependent institutional arrangements, intergovernmental authority asymmetries and capacity gaps shape the conditions for urban climate adaptation in Kathmandu Valley. These expectations structure data collection and guide the initial coding list of the empirical materials. The study also merges inductive methods as it introduces the needed flexibility as in the context of Nepalese federalism and local climate governance which helps identifying empirically complex information that may not be fully captured by existing theory. This allows for the analysis to remain open to unexpected patterns and meaning that emerge from the data itself (*Saunders & Darabi, 2024*). The structured deductive approach supports theoretically anchored but empirically responsive study.

4.3 Research strategy

4.3.1 Single-Case Study Design

The study employs a single-case, qualitative design, with urban climate-adaptation governance in Kathmandu Valley as the case. A single-case design is appropriate here because the aim is not statistical generalization but analytical depth; to trace how institutional dynamics unfold in a

specific, theoretically significant context (*Yin, 2018*). Kathmandu is treated as a bounded governance space in which federal, provincial and municipal authority layers are simultaneously operative and where the tension between formal institutional reform and functional capacity constraints are especially visible. As Nepal's primary urban agglomeration and political administrative center, the city concentrates the actors, agencies and policy processes through which federal climate adaptation mandates are meant to be operationalized making it the most appropriate site for examining the research questions.

Kathmandu Valley comprises three districts—Kathmandu, Lalitpur, and Bhaktapur and covers approximately 899 square kilometers, with a population density of 2,793 people per square kilometer. The valley is located between 27°38'32"–27°45'7" N latitude and 85°16'5"–85°22'32" E longitude at an average elevation of approximately 1,350 meters above sea level. It experiences a mild temperate climate with an average annual temperature of 16.1°C, average humidity of approximately 75%, and annual precipitation of around 2,812 mm, with heavier rainfall during summers than winter influenced by the monsoon winds (*Pantha et al., 2024*). Nepal has six metropolitan cities in total, two of which Kathmandu Metropolitan City (KMC) and Lalitpur Metropolitan City (LMC) are located within the valley. KMC has a population of approximately 2,041,587, covers 50.67 km² across 11 municipalities and 32 wards. LMC covers 37.4 km² and comprises 6 municipalities across 29 wards.

Empirically, the study focuses primarily on KMC, where data collection is most concentrated. LMC is included within the broader case, though only few interviews were conducted there. No interviews were conducted in Bhaktapur, which does not hold metropolitan-city status. This uneven coverage reflects practical access constraints rather than a deliberate comparative design and is acknowledged as a limitation: the analysis of local-level dynamics is strongest for KMC, while Lalitpur and Bhaktapur contribute primarily as contextual settings. Other emerging Nepali cities such as Pokhara and Bharatpur are not included as empirical cases, in order to maintain analytical depth rather than pursue a broader comparative study.

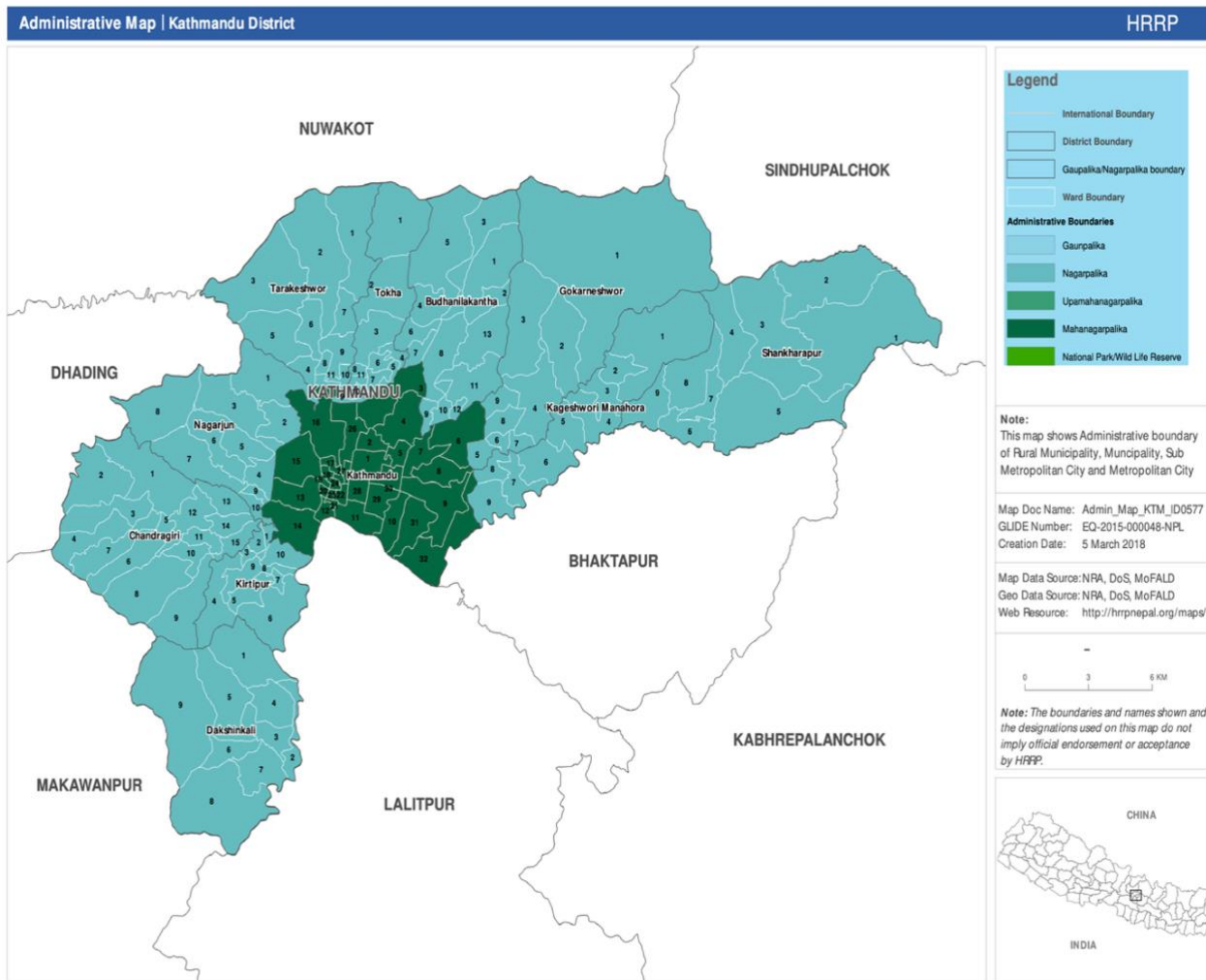


Figure 2 Administrative boundary of Kathmandu metropolitan city (KMC) (*study area: dark green*)

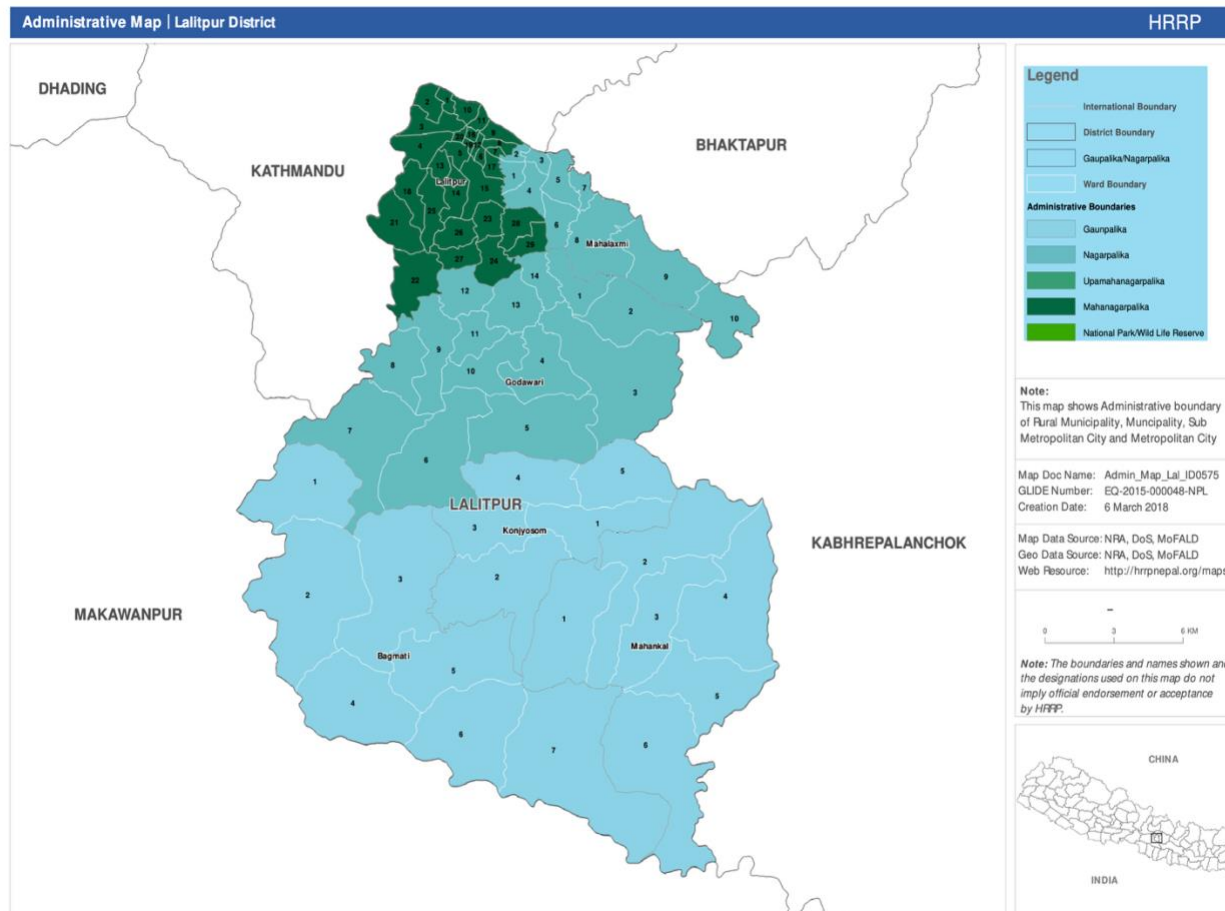


Figure 3 Administrative boundary of Lalitpur metropolitan city (LMC) Study area: dark green source HRRP²

4.4 Research Design

The study employs a descriptive-exploratory research design to systematically analyze why institutional constraints on urban climate adaptation in Kathmandu Valley persist despite federal reforms and decentralization. The descriptive dimension maps how key actors perceive governance structures, roles, mandates, and capacity gaps in climate-change adaptation. The exploratory dimension investigates the historical, multilevel, and institutional mechanisms that reproduce these constraints over time.

The research process unfolds across three stages. In the first stage, a conceptual framework drawing on HI, MLG and ACW is developed to guide data collection and analysis (Chapter 3). In the second

² https://hrrpnepal.org/uploads/media/eVfOFb5R4luy7E29B3vH_2018_03_06.pdf

stage, empirical data are collected through semi-structured expert interviews (n=18) and qualitative questionnaires (n=6) with stakeholders involved in urban climate-adaptation governance in Kathmandu Valley. In the third stage, interview and questionnaire data are coded and interpreted using MAXQDA software, guided by the three research questions, to trace patterns of institutional persistence, multilevel asymmetries, and capacity reproduction. In the final stage the data collected through interview and questionnaires are triangulated through major adaptation policies through desk work. These stages are illustrated in figure 5.

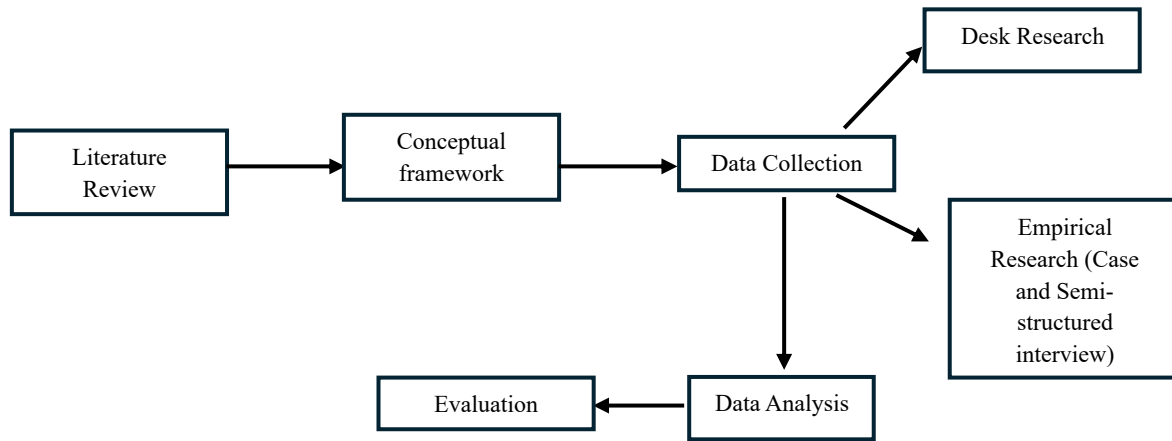


Figure 4 Research Design

4.5 Data Collection

Primary data were collected through 18 semi-structured expert interviews and 6 qualitative questionnaire responses, conducted between August, September and November 2025 with stakeholders involved in urban climate-adaptation governance in Kathmandu Valley. Some of the interviews were conducted online in August, and in-person interviews took place in Kathmandu in September and November. Key informants were initially identified through internet searches of organizations with current institutional involvement in climate governance, including government offices, climate-focused NGOs, and community-engagement initiatives. They were contacted via email and, where necessary, followed up by phone to schedule interviews. In-person interviews were conducted at respondents' workplaces in Kathmandu.

Fieldwork was partially disrupted by a nationwide curfew imposed in response to widespread civil protests, which temporarily restricted access to some respondents. Several interviews took place during this period of civil unrest, when parts of Kathmandu were intermittently inaccessible. This context introduced additional sensitivities: some government respondents were cautious about the content of their accounts, and access to certain ministries was limited during curfew days. Despite this constraint, the study achieved a sufficient range of perspectives across governance levels and institutional types.

All interviews were audio-recorded with participants' consent, transcribed verbatim, and anonymized using the coding system shown in Table 1. The qualitative questionnaire was administered to key respondents who could not be reached for in person interviews in Kathmandu, using the same topic guide to ensure thematic consistency. Of the twenty-two questionnaire responses received, six were fully completed and only those were included in the analysis. Thematic sufficiency was reached organically during the data collection process. By approximately the fifteenth interview, recurring themes including fiscal dependency, limited local autonomy, fragmented intergovernmental coordination, and capacity constraints were appearing consistently across respondents with no substantively new dimensions emerging. The final three interviews served primarily a confirmatory function. This pattern of thematic repetition across diverse stakeholder positions government, NGO, INGO, and research provides confidence that the empirical material captures the core institutional dynamics of urban climate-adaptation governance in Kathmandu Valley. In addition to primary data, the study draws on desk research of key national climate-change policies and legal frameworks, including the National Climate Change Policy NCCP (2019), the Local Governance Operation Act LGOA (2017), and the National Adaptation Plan (NAP) (2021–2050). These documents are used to contextualize institutional reforms and trace the evolution of formal mandates for urban climate adaptation and are treated as background material rather than primary empirical data.

4.5.1 Sampling Rationale

Informants were selected through purposive sampling, guided by two primary criteria: institutional position and direct involvement in urban climate-adaptation governance in Kathmandu Valley. The sampling strategy was designed to capture perspectives across governance levels and organizational types, combining government actors including municipal officials, planners, and ministry-level

civil servants with representatives of NGOs and INGOs actively engaged in climate-adaptation programming at the local level. A minimum threshold of direct institutional involvement in climate governance was applied when selecting public-sector informants, ensuring that interviewees held roles with substantive decision-making or implementation responsibilities. For NGO and researcher informants, the criterion was organizational engagement with local-level climate adaptation rather than a specific seniority threshold, reflecting the diverse ways in which non-state actors contribute to governance in this context. Initial contacts were identified through targeted internet searches of relevant institutions and organizations, and in several cases a degree of snowball sampling was employed, where interviewees suggested additional contacts within their professional networks. This was not systematic across all cases but was used pragmatically where access to specific governance levels or institutional roles was otherwise difficult to secure.

4.5.2 Questionnaire Justification

The decision to supplement interviews with a qualitative questionnaire was primarily pragmatic. Government officials and senior institutional actors proved difficult to access during fieldwork, resulting in a smaller number of interviews with public-sector respondents than originally planned. To broaden the empirical base and capture perspectives from additional stakeholders — particularly ward level officers whose experiential knowledge is valuable even where their seniority or policy involvement did not meet the threshold applied to expert interviewees. The qualitative questionnaires were sent to their email addresses. The questionnaire is therefore treated as supplementary rather than equivalent to interview data, and responses are analyzed alongside but distinguished from interview material in the thematic coding process.

4.5.3 Interview design

Interviews lasted between 45 and 90 minutes and followed a topic guide structured around the three research questions: the evolution of roles, mandates, and practices over time; constraints on local autonomy within the federal and multilevel governance system; and the reproduction of institutional capacity gaps. Questions probed actors' experiences of historical trajectories, intergovernmental relations, fiscal dependencies, institutional memory, and organizational routines distinguishing between organizations (municipalities, ministries) and institutions (rules, norms, procedures). The description of the expert interviewees, respondents from qualitative questionnaire and their coding process is listed in Table 1 and 2. A sample questionnaire is provided in Appendix 1

Table 1 Overview of expert interviewees (in-person and online)

S.N.	Actor Type	Affiliation	Designation	Code
1	Researcher	Independent	Researcher	RES_01_18:08:2025
2	Researcher	Forest Action Nepal	Researcher	RES_02_26:08:2025
3	INGO	NDC Partnership	In-country facilitator for NDC Partnership	FAC_01_20:08:2025
4	INGO	Worldwide Fund (WWF, Nepal)	Head of Climate and Energy Program	INGO_01_19:08:2025
5	INGO	Oxfam	Strategic Advisor	INGO_02_17:09:2025
6	INGO	Practical Action Nepal	Adaptation Research Lead	INGO_03_18:09:2025
7	INGO	Practical Action Nepal	Researcher	INGO_04_24:09:2025
8	Gov	Nepal Climate Change Knowledge Management Center	Government scientific officer	h) Gov_01_05:09:2025
9	Gov	Kathmandu Metropolitan City Office Env. Division	Head Engineer	Gov_02_04:09:2025
10	Gov	Ministry of Forest and Environment – Climate Change Management Department	Policy Officer	Gov_03_13:09:2025
11	Gov	Ministry of Forest and Environment – Climate Change Management Department	Ex-Department Chief	GOV_04_15:09:2025
12	Gov	Lalitpur Metropolitan City	Engineer	GOV_05_15:09:2025
13	Gov	UNEP, Urban EBA project at Kathmandu Valley Development Authority (KDVA)	Head of project department	GOV_06_08.09.2025
14	NGO	ALL IN Foundation	Chief Executive Officer	NGO_01_15:09:2025
15	NGO	Nepalese Youth for Climate Action (NYCA)	Board Member	NGO_02_18:09:2025
16	NGO	Bignaharta Nepal	Director	NGO_03_22:09:2025
17	NGO	YASEN	Founder	NGO_04_23:09:2025
18	NGO	Urban Green Building Nepal	Founder	NGO_05_19:09:2025

Table 2 Overview of qualitative survey questionnaires

S.N.	Actor Type	Affiliation	Designation	Code
1	Researcher	/	/	QAL_RES_01_2026
2	INGO	/	/	QAL_INGO_02_2026
3	Gov	Local level	/	QAL_GOV_03_2026
4	Gov	Local level	/	QAL_GOV_04_2026
5	Gov	National level	/	QAL_GOV_05_2026
6	Researcher	/	/	QAL_RES_06_2026

4.6 Positionality

Since the study treats knowledge as socially produced and institutionally situated, it is necessary to acknowledge my own position within the research, how it may have shaped what was asked, what was heard and the way findings were presented.

I grew up in Kathmandu Nepal, which gives me a unique and direct familiarity with the governance culture, bureaucratic practices, and political dynamics this thesis examines. I conducted interviews in Nepali and English according to respondents' preferences. This familiarity encouraged a degree of honesty among some respondents, particularly government officials, who recognized that I understood the significance of what they were describing and the informal institutional practices they were referencing. It also meant that I brought into the research a prior understanding of how Nepal's governance system constrains local governments institutionally rather than through individual failures – a framing that aligns with the theoretical framework and shaped what I found credible to present in the thesis. This was controlled by framing closest to what interviewees meant in their own language and by triangulation across stakeholders who held structurally diverse positions in the governance system.

My academic writings and training in western institutional frameworks have also shaped the conceptual lexicon. Frameworks used in the thesis such as historical institutionalism, multi-level governance and adaptive capacity carry their own assumptions on how institutions should work,

what empowerment means and what constitute a governance failure that might not completely align with Nepali realities. I have been attentive to moments where evidence showed more than what theory predicted, for example: critical junctures failing to produce institutional change or cases from LMC that diverged from the structural patterns dominant in KMC and have shown them in data presentation sections.

4.7 Data Analysis

The empirical material from semi-structured interviews and qualitative questionnaires is analyzed through thematic content analysis, organized and coded using MAXQDA software. To strengthen the credibility of the findings, the study employs source triangulation across two dimensions. First, accounts are compared across stakeholder categories particularly between government officials and researchers or NGO and INGO representatives to identify convergences and divergences in how institutional constraints, resource flows, and governance practices are perceived and described. Second, interview and questionnaire accounts are cross-checked against the desk research of national climate-change policies and legal frameworks to assess the extent to which actors' narratives correspond to, diverge from, or reinterpret formal institutional mandates. This triangulation across sources aims to map the gap between formal institutional design and experienced governance reality which is central to the research question.

Data analysis follows a deductive procedure. A deductive codebook, developed from the three components of the conceptual framework, historical trajectories and institutional persistence, multilevel governance asymmetries, and institutional and adaptive capacity is applied in the first pass to tag theoretically relevant segments. Codes are then refined inductively as new patterns emerge from the data, allowing the analysis to remain responsive to actors' own framings and experiences. Together, the codebook is organized around 10 parent codes, each corresponding to a core analytical dimension of urban climate-adaptation governance. Descriptively the parent codes consist of the following headings: *Parent Code 1*: Institutional Role and Mandates capture how institutions understand their responsibilities, legitimacy, and decision space, including formal mandates, role evolution over time, and perceived authority. *Parent Code 2*: Historical Institutional Dynamics traces path dependence, critical junctures such as the 2015 earthquake and federal restructuring, and forms of incremental change including layering and drift showing how past decisions and legacies shape current governance. *Parent Code 3*: Actor Influence and Power

Dynamics examine who shapes decisions—internal actors (mayors, ward chairs, departments), external actors (donors, NGOs, consultants), and communities—and how multi-level coordination or friction affects outcomes. *Parent Code 4: Resources & Capacity* focuses on financial resources, human resources, technical capacity, and changes in capacity over time. *Parent Code 5: Autonomy and Local Empowerment* analyze financial, administrative, and political autonomy, as well as constraints from federal or provincial interference and the freedom to innovate. *Parent Code 6: Knowledge and Learning Systems* traces monitoring and evaluation practices, single- and double-loop learning, institutional memory, and internal communication. *Parent Code 7: Information Access and Use* examines access to climate and risk data, data quality, and the gap between data availability and actual use in decision making. *Parent Code 8: Variety and Diversity of Solutions* capture multi-actor engagement, multi-sector coordination, and the diversity of adaptation options (e.g., engineering, ecosystem-based adaptation, community-based approaches). *Parent Code 9 and 10: Barriers and enablers to Effective Adaptation* cross-cutting obstacles and facilitating factors across political, financial, coordination, knowledge, and regulatory dimensions. The full codebook, including all subcodes, is provided in the Appendix 2.

The analysis proceeds in several steps. Transcripts and questionnaire responses are first read in full to gain an overall sense of the material. The deductive codebook is then applied to tag relevant segments, after which codes are refined iteratively, and representative quotes are selected to illustrate key themes. Questionnaire responses are coded using the same thematic structure as interviews, allowing comparison of patterns across data collection modes.

4.8 Ethical considerations

The study adheres to standard ethical principles for social-science research. All participants were informed about the purpose of the study, their right to withdraw at any time, and how their data would be used and stored, including in the initial scheduling communication. Informed consent was obtained before each interview began, and audio recordings were made only with explicit permission. Data are stored securely and are accessible only to the researcher and, where required, the thesis supervisor.

4.9 Limitations

The study has several limitations that should be acknowledged transparently. First, the single-case focus on Kathmandu Valley limits direct generalizability to other Nepali cities or growing cities in the global south with comparable federal settings. While the institutional mechanisms identified may carry broader theoretical relevance, the empirical findings are context-specific and should not be assumed to apply automatically elsewhere. Second, empirical coverage within the Valley is uneven: data collection is concentrated in KMC, with only a small number of interviews in LMC and none in Bhaktapur. Third, the study relies primarily on the perspectives of officials, planners, NGO representatives, and donors, which may underrepresent the views of marginalized communities, frontline implementers, and ordinary citizens. Fourth, the analysis is based on a cross-sectional account of governance dynamics spanning 2015 to 2025, reconstructed through actors' narratives and policy documents rather than direct longitudinal observation. Fifth, desk research on national policies and legal frameworks is used for contextualization only and is not systematically coded or compared as a primary source of empirical data. These limitations are inherent to the chosen design and point towards future research using multi-city designs, longitudinal or mixed methods approaches for data collection.

Chapter 8: Discussion

Urban climate adaptation governance in Nepal presents a specific analytical conundrum: formal authority has expanded, policy frameworks have multiplied, and motivated actors exist at multiple levels yet municipalities in Kathmandu remain functionally constrained in governing climate change adaptation. Chapter 8 develops the theoretical explanation for this conundrum by showing how the three analytical frameworks applied in this study historical institutionalism, multi-level governance, and the adaptive capacity wheel converge on a single structural account of why constraints persist, how they reproduce, and what this means for existing theory.

8.1 Path Dependence, Layering, Stickiness and Drift: How Constraints Reproduce

The first research question asked how institutional roles, mandates and practices related to climate adaptation have evolved over time in Nepal's urban governance context. The findings show that the evolution has been formal rather than functional. Roles expanded on paper through LGOA 2017 and NCCP 2019, while the roles that operate in everyday governance remained shaped by trajectories established well before federalization.

Four mechanisms of institutional reproduction explain the pattern, they show how constraints persist but also how they actively reproduce themselves through everyday governance practices. *Path dependence* operates through the donor-driven origins of Nepal's climate governance: early NGO-led pilots created project-based interventions that were never truly institutionalized within the formal state systems, leaving municipalities embedded in reactive donor cycles even after federalization formally changed their authority (as mentioned in Ch 5.5). *Layering* operates through the 2015 federal transition itself: new institutions and mandates were placed on top of older centralized structures, creating a hybrid arrangement rather than a complete overhaul, preserving old power asymmetries beneath new formal frameworks. Because the reforms added mandates without dismantling the fiscal and administrative channels through which central authority had always operated. *Institutional stickiness* operates through informal knowledge transfer between predecessor and successors which are inherited practices that kept continuing not because they were progressive, created positive impact or evaluated for their effectiveness. But because routines of forestry-centric environmental governance, waste-management prioritization, and engineering-dominant adaptation solutions are transmitted independently of whether the formal mandate has

changed. *Drift* operates through selective LAPA deployment and the reorientation of national adaptation priorities toward loss and damage funding: even though the policy has formally remained unchanged, operational meaning have been redirected towards easier contexts and towards priorities where international finances flow. These four mechanisms explain the explanatory logic that is absent from existing Nepal literature explaining that constraints exist but how they actively reproduce themselves through ordinary functioning of governance.

One mechanism, institutional stickiness operating through informal predecessor-to-successor knowledge transfer was not fully anticipated by the theoretical framework. Historical institutionalism's concept of institutional stickiness typically operates at the level of formal rules, organizational cultures, and policy instruments that resist displacement. The empirical material revealed that bureaucratic routines are transmitted through informal channels between outgoing and incoming officials independently of whether the formal mandate governing those routines has changed. An official inherits not only a position but a set of unwritten practices such as ways of handling climate relevant requests, which agencies to defer to, which actions to avoid and these behaviors persist because they are passed on from one person to another. This adds a layer to how reproduction operates beyond the formal that existing HI accounts do not fully capture.

This evidence also challenges Capoccia's (2016a) optimism regarding transformative junctures. Federal restructuring, the 2015 earthquake, the 2017 Terai floods, and the 2024 Kathmandu floods all functioned as awareness moments rather than institutional turning points. Bureaucratic inertia, political instability, and continued federal dominance converted potential windows of change into reinforcement of maladaptive trajectories. This confirms that critical junctures do not produce transformation when the enabling conditions fiscal autonomy, administrative readiness, and political space to act remain controlled at higher governance tiers.

8.2 Formal vs Symbolic Decentralization

The second research question asked why formal decentralization fails to produce functional autonomy for urban local governments. The findings show that the failure is not simply a byproduct of poor implementation or insufficient capacity; it is structurally encoded in the legal framework governing local governments.

The 2015 Constitution and subsequent LGOA 2017 formally redistributed policymaking and budgetary authority to local governments, but substantive control remains vested in federal and

provincial tiers through entrenched financial, administrative, and political mechanisms. Local budgets require federal reimbursement via District Treasury Offices through a three-modal budgetary system. International climate finance flows through federally controlled channels with no independent local access pathway. Staffing is routed through central personnel mechanisms. Implementing external agencies such as I/NGOs, community initiatives that are theoretically there to support local governance in adaptive practice demand legal justification for any innovation, discouraging the experimentation that adaptive governance requires.

This results in a condition where the formal structure of multi-level governance exists while substantive authority remains concentrated at the center – reflecting only as a symbolic decentralization. It is produced by path-dependent concentration of power operating through layering, maintained by the fiscal and administrative mechanisms that federalism reformed on paper but not in practice. Multiple evidence of these have been explained in detail in Chapters 5 and 6. Unlike genuine polycentricity, which enables context-specific experimentation and scale-appropriate authority allocation, Nepal’s federal arrangement creates the appearance of local empowerment while preserving the functional conditions of central dependence. Municipalities hold statutory mandates for planning and implementation while remaining unable to access climate finance independently, recruit specialized staffing, or implement programs without approval chains that can delay action by years.

The limited success cases documented in section 6.4 the NDRRMA, KVDA land pooling, and biodiversity implementation confirm the structural logic by contrast: where mandate specificity and dedicated departmental authority exist, adaptive governance follows. This extends Ojha et al.’s (2014) findings on formal participatory reforms failing to deliver substantive empowerment into the specific domain of post-2017 climate governance under federalism, where new formal authority has not altered the substantive distribution of power.

8.3 Structural Reproduction of Adaptive Capacity Constraints

The third research question asked how key dimensions of institutional capacity shape adaptive governance in urban municipalities and why similar capacity gaps persist over time. The findings show that technical, financial, and knowledge capacity do not operate as independent variables that can be improved in isolation. They form a mutually reinforcing system, a governance trap in which each dimension’s weakness actively reproduces weakness in the others.

Insufficient and inaccessible finance and structural limitations outlined in mandates prevents municipalities from recruiting and retaining specialist technical staff. The absence of specialist staff means that climate knowledge generated through projects cannot be absorbed into institutional routines and departs with individual staff when they are transferred. The absence of embedded institutional memory means municipalities cannot demonstrate their resource needs or capacity autonomously, as a result they will be underrepresented or excluded from financial decision making that would address those needs. The cycle then begins again. This governance trap is defined as mutually reinforcing lock-ins where weak financial flows and structural limitations limit technical expertise, which undermines implementation, which then erodes learning and creates fragility. This framing shifts the interpretation from ad hoc deficits that can be solved by targeted training or budget transfers as identified by much prior research to rather a structural problem requiring institutional redesign. This idea of governance trap where institutions influence not only the individual choices but shape the collective capacities that reproduce over time as identified by *(Lockwood et al., 2016)*.

A finding that emerged beyond what the ACW framework anticipated is the mutually reinforcing nature of these three dimensions. The ACW treats variety, learning capacity, and availability of resources as analytically separable dimensions that can in principle be assessed and addressed independently *(Gupta et al., 2010)*. The empirical material from Kathmandu Valley complicates this assumption: interventions targeting any single capacity dimension in isolation through training programs, budget transfers, or knowledge management systems cannot generate sustainable change unless the structural conditions linking all three dimensions are simultaneously addressed. Adaptive capacity thus emerges as an institutionally embedded relational property rather than a discrete variable that can be measured and filled. This reconceptualization explains why policy abundance and motivated actors can coexist in fragmented urban adaptation settings such as in Kathmandu without any functional transformation. It also explains why linear models of capacity building which assume that improved information, funding, or expertise will automatically lead to implementation, consistently fail to explain the persistence of governance dysfunction.

8.4 Integrating the Three Frameworks

The three analytical lenses converge on a single structural explanation: urban climate governance constraints in Nepal are produced through the interaction of path-dependent institutional

reproduction, hierarchically constrained decentralization, and structurally embedded capacity deficits and they reinforce one another recursively.

Path-dependent centralization through informal legacies and donor cycles limits meaningful decentralization under federalism. Constrained autonomy through staff transfer rates, inability to access finances independently, and intergovernmental gatekeeping prevents municipalities from building stable technical and financial capacity. Weak capacity in turn reinforces dependence on higher governance tiers and donor systems, reproducing the original institutional trajectory. This study contributes to three scholarly debates. First, it extends historical institutionalism by demonstrating how critical junctures in federal climate governance may be absorbed into path-dependent reproduction rather than generating transformation, challenging *Capoccia's (2016a)* optimism and showing that the informal reproduction mechanisms of stickiness and drift can outweigh the formal reorganization of a critical juncture. Second, it contributes to multi-level governance theory by showing that formal decentralization without fiscal and administrative autonomy produces symbolic devolution and limits urban adaptive experimentation. *Ojha et al. (2014)* provides evidence of this in the post-2017 climate governance context where new formal authority has not altered the substantive distribution of power in the country. Third, it reconceptualizes adaptive capacity as a historically embedded institutional property rather than merely a function of skills, finance, or awareness (*Gupta et al., 2010*), explaining why similar resource endowments produce different outcomes in different institutional contexts and why the governance trap cannot be resolved through any single dimension intervention.

8.5 Implications for Urban Climate Governance in Nepal

The findings suggest that strengthening adaptive governance in Kathmandu requires institutional transformation rather than additional policy production or short-term projects. Each implication that follows responds directly to a specific mechanism of institutional reproduction identified in the empirical chapters.

The most immediate implication concerns mandate clarity and accountability mechanisms. Because ambiguity in mandates is structurally encoded within LGOA 2017 through concurrent jurisdiction provisions, these are aspirational in nature rather than binding climate adaptation commitments. To limit the condition of having too many responsibilities and too little clarity on who does what requires secondary legislation or intergovernmental agreements that specify how shared authority

is exercised in practice, legally bounded climate targets and compliance consequences for adaptation planning at the local level. Without this, the pattern established in chapters 5.2 and 5.3 where accountability will remain aspirational and municipalities will keep promising the production of plans with no meaningful climate impact. Without jurisdictional consequences or clarity these tendencies will persist across future planning cycles.

The second implication concerns the climate fiscal architecture prevalent in Nepal. Because of financial centralization reproduced through LMBIS, the ways in which conditional grants operate and the gatekeeping of international climate finance at the federal level, devolving fiscal authority to the local level requires redesigning the overall process of financial decision making. This needs to be done together with the local levels where they become autonomous implementers rather than a mere expenditure unit spending whatever is allocated by the federal level. It requires detangling the access channels where 80 percent of the climate fund reaches the local level not just promises and in a sustainable way that they can generate their own climate funds in the future. And finally treating approvals as an accountability measurement not a mandatory central supervision. Local governments need a direct pathway to climate finance that does not require federal ministerial consent at every step, and the tri-modal budgetary system needs simplification that gives municipal accounts real cash rather than accounting entries.

The third implication concerns dedicated municipal climate institutions or units. (*Aligica & Tarko, 2012*) through their expansion of Ostrom provide evidence that “*polycentric arrangements with small, medium and large departmental systems generally outperformed cities that had only one or two large departments*” (p.243). Overall climate and environmental issues ranging from mitigation, adaptation, biodiversity, forest, pollution control and waste management have been compressed as the responsibility of larger Ministry of Forest and Environment and or sub-unit Climate Change Management Divisions. This overall diffusion of climate functions into either disaster or forestry departments with no special mandate reproduces institutional stickiness. The structural response would be the establishment of dedicated municipal climate units with their own decision-making authority, technical staffing lines, and budget allocation not as focal points, not as coordination nodes active only during a climate crisis, but as empowered functional departments. The evidence from Lalitpur and from the NDRRMA and KDA successes documented in chapter 6.4 confirms that

mandate specificity and clear departmental ownership are the conditions under which adaptive governance becomes possible even within broader structural constraints.

The fourth implication concerns knowledge management and institutional memory. Because weak institutional memory is reproduced as there is absence of embedded documentation systems, transfer mechanism and the exclusion of the local level from the NAP's monitoring and reporting sequencing. Addressing this vacuum requires formalized knowledge transfer obligations as part of staff rotation procedures, investment in locally operated data management infrastructure, and inclusion of municipal governments in the monitoring and evaluation planning from the first phase rather than as a future expansion stage.

The fifth implication concerns the shift from donor-driven projectization toward domestically institutionalized planning. Because path dependence is reproduced through the early establishment of NGO-led, donor-funded climate governance that municipalities were never required to absorb and institutionalize, reversing this trajectory requires domestic budget allocation for climate planning functions as a recurrent cost rather than a capital project. Formal mechanisms for transferring the technical knowledge that has accumulated in NGOs and INGOs into municipal institutional systems would benefit greatly rather than leaving it to circulate informally through staff mobility.

8.6 Limitations and Scope

While the analysis provides institutional insight grounded in qualitative evidence from selected urban contexts and stakeholder perspectives, several limitations shape the scope of the research's conclusions. The study is concentrated on Kathmandu, which is Nepal's capital and the most donor-intensive urban context, making patterns of federal dominance, donor dependence, and nominal polycentricity particularly visible. But these underlying conditions may overstate their intensity relative to smaller or more peripheral rural municipalities away from Kathmandu. The interview samples overrepresent federal level officials, INGOs, and NGOs while under-presenting ward-level officials and informal settlers who have been exposed to adaptation gaps. The qualitative design traces mechanisms and explanatory pathways but cannot quantify the prevalence of the patterns identified or establish whether those identified mechanisms operate uniformly across different types of municipalities. Institutional dynamics are also not static, evolving political or fiscal reforms could alter the trajectories observed at the time of data collection. The analytical conclusions are

therefore contextually grounded and not predictive in nature at all. Future research using comparative municipal case studies or longitudinal designs would strengthen the generalizability of the mechanisms identified in this research.

Chapter 9 Conclusions

Urban climate adaptation in Nepal does not fail or lack policies, frameworks or motivated actors. Despite substantive progress at the policy level, adaptation governance is failing because of how institutional arrangements organize the conditions for action — how authority is distributed, how resources flow, how knowledge accumulates, and how practices are reproduced across planning cycles. Collectively, these arrangements systematically prevent local governments from acting on the mandates they formally hold. The research therefore is a way to explain why this is so, and why such constraints persist across decades of policy evolution, federal restructuring, and formal expansion of local authority.

The answer lies in the interaction of three interconnected dynamics. First, Nepal's governance of climate adaptation is path dependent. Early NGO and donor driven initiatives established project based reactive logic that was never really absorbed by the state. Federal restructuring in 2015 did not sever this path dependence, instead it layered new mandates onto an unreformed foundation, creating a hybrid institutional governance structure. This expanded the formal responsibilities while leaving fiscal dependencies, inherited bureaucratic norms and donor dependence intact creating stickiness institutions could not sever. As more international debate is shifting towards loss and damage, national adaptation priorities are also subtly drifting there, not as a necessity for the local level but for finding the money trail to finance ambitious climate commitments. Critical junctures including the 2015 earthquake, the 2017 Terai floods, and the 2024 Kathmandu floods generated awareness but not structural transformation, because the conditions necessary to convert shocks into new institutional trajectories: devoted political will, resource autonomy, and local administrative readiness, remained absent at the local level.

Second, formal decentralization in Nepal has produced a governance structure that resembles multi-level authority distribution on paper while decision making, fiscal control and administrative discretion concentrate mainly at the federal level in practice. The LGOA 2017 and NCCP 2019 assign municipalities with broad climate responsibilities while encoding dependency to the center in the same documents through conditional grants reimbursed after multiple administrative layers, mandatory federal consent for implementation of external partnership and staffing allocation controlled through central personnel management system. The result is that the local government remains structurally subordinate even where they hold formal authority, unable to access

international climate finance independently, recruit specialized staffing or implement programs without federal approval chains. This can substantially delay climate adaptation programs by years, in a time when it is already known that adaptive governance is not able to catch up fast enough to the cascading effects of climate change.

Third, the institutional capacity required for adaptive governance cannot accumulate efficiently under these conditions. Technical capacity is diffused across generalist disaster departments without specific climate units at the local level and dedicated climate mandates catered towards the need of the urban local level. Financial resources exist only in allocation terms as capital budgets hover around sixty percent utilization thresholds and most returned unused due to procurement rigidities and disbursement bottle necks. Climate knowledge has remained personalized rather than institutionalized or embedded within the system, most knowledge leaving the institution with staff transfer. Monitoring and evaluation frameworks at the national level sequence local governments last even if municipalities are the first point of contact for data and effective climate program implementation. These do not operate in silos, instead they form a mutually reinforcing governance trap in which weak finance limits specialized staffing, weak staffing yields minimal experiential learning, and absent institutional memory reproduces the same capacity gaps across successive planning cycles regardless of what resources and plans are made available.

Theoretically, the thesis makes three contributions. First, it extends historical institutionalism by showing that critical junctures in federal climate governance can be absorbed into path-dependent reproduction through layering, drift and institutional stickiness. The study adds transmission of bureaucratic routines and knowledge from person to person independently of formal mandate changes. Second, it contributes to multi-level governance theory by demonstrating that formally polycentric authority does not produce functionally polycentric governance when the fiscal, administrative, and knowledge systems through which that authority operates remain centralized. This produces the appearance of local empowerment while preserving the structural conditions of central dependence. Third, it reconceptualizes adaptive capacity as a historically embedded institutional property shaped by governance arrangements, power relations, and inherited routines, rather than a discrete stock of skills or resources transferable through training or budget allocation. This reconceptualization explains why similar resource endowments produce different outcomes

across institutional contexts, and why linear models of capacity building consistently fail to generate sustainable adaptive governance in cities like Kathmandu.

Strengthening urban climate governance in Nepal requires confronting these structural arrangements rather than adding further policy layers. Mandate ambiguity must be resolved through binding secondary legislation rather than concurrent jurisdiction provisions that assign responsibilities to all levels equally without accountability mechanisms. Fiscal structures must be redesigned to give municipalities direct access to climate resources rather than downstream reimbursement of centrally managed funds. This is achievable through dedicated municipal climate budget lines and simplified intergovernmental transfer mechanisms. Dedicated municipal climate units with specialized staffing and independent decision-making authority must replace the current model of climate functions nested within disaster departments. Knowledge transfer must be formalized as an obligation within staff rotation procedures, and local governments must be included in monitoring and evaluation planning from the initiation stage and not as an afterthought.

The study is necessarily bounded. Its empirical focus is on Kathmandu Valley; it relies on reconstruction of stakeholder narratives and centers governance actors rather than frontline communities or ward-level implementers. Comparative multi-city research, longitudinal process tracing, and community-level perspectives are needed to extend and test the mechanisms identified here. What the thesis establishes is that urban climate adaptation in Nepal is not an implementation problem or a policy-making problem — it is a structural problem produced by historically embedded governance arrangements, hierarchically constrained decentralization, and institutionally reproduced capacity deficits that reinforce one another in ways that formal reform alone has been unable to disrupt. Understanding these necessary conditions is the starting point for any governance change, more than adding another layer of already oversaturated and overburdened institutional landscape.

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